

Setting Up a New Project



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Why Tooling is Needed

Starting point

Transform JSX to JavaScript

Process JavaScript files

Run a development server

Automatically update browser

Create a production build



Options for Tooling

Do It Yourself

Use a ready-to-go development environment

- Vite
- Next.js



Install Node.js

<https://nodejs.org/en/download>



SWC

Speedy Web Compiler

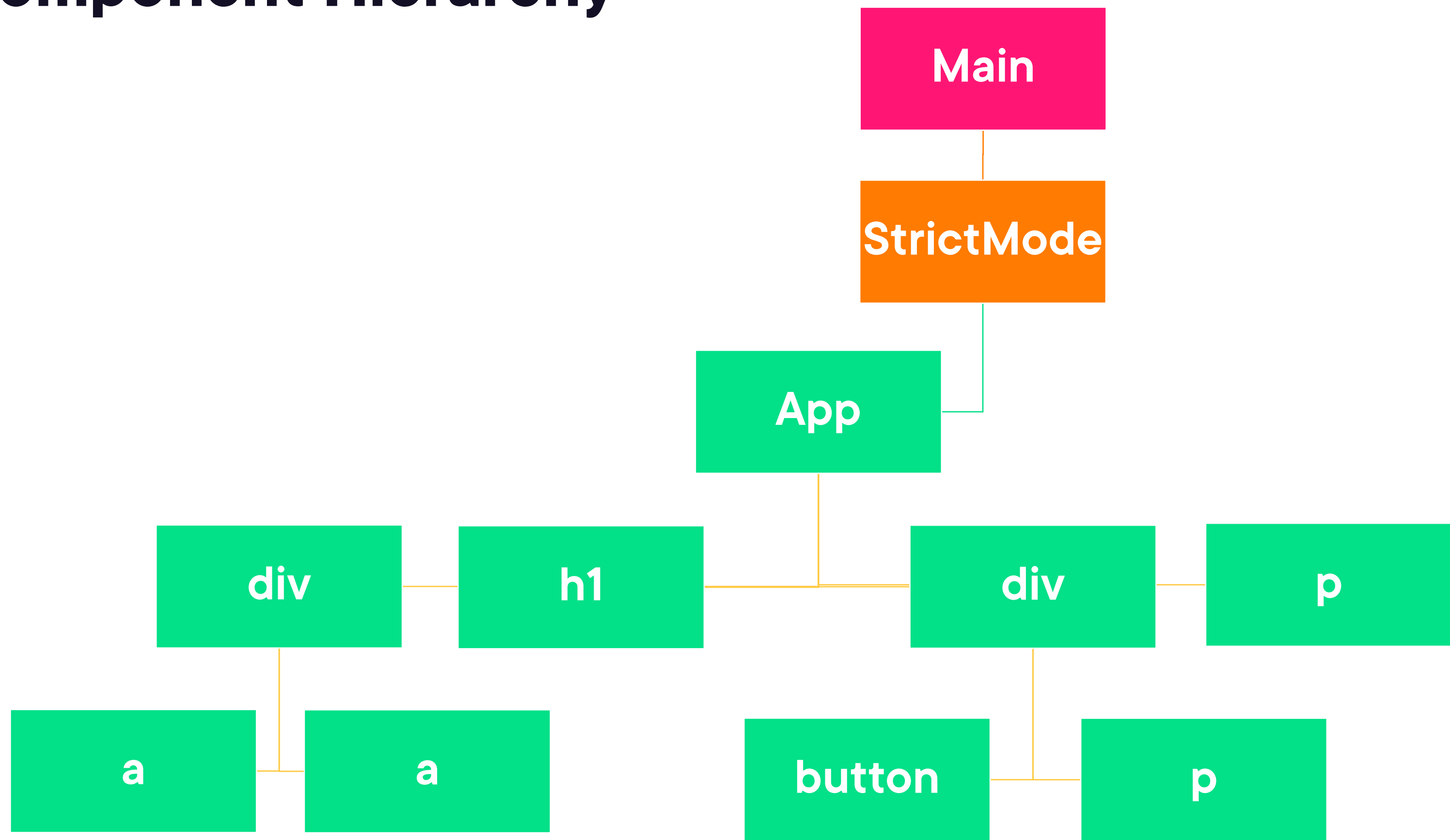
Transpiles JSX to JavaScript

Performance

Update browser on the fly



Component Hierarchy



Modules

Components are in modules

JavaScript files with import and/or export statements

Not specific to React

Vite relies on them



Modules

module.js

```
const doSomething = () => {  
  ...  
};  
export { doSomething };
```

anotherModule.js

```
import { doSomething } from "./module";  
  
doSomething();
```



Modules

module.js

```
const doSomething = () => {  
  ...  
};  
export default doSomething;
```

anotherModule.js

```
import do from "./module";  
  
do();
```



React and Modules

Banner.jsx

```
const Banner = () => (  
  <h1>  
    This is a banner  
  </h1>  
>;  
export default Banner;
```

Greeting.jsx

```
import Banner from "../Banner";  
  
const Greeting = () => (  
  <Banner />  
>;  
export default Greeting;
```



Reasons to Use Modules

Code structure

Reusability

Encapsulation

Performance

Scalability

Easier debugging and testing

Collaboration

Needed for bundling



Running the Finished Project

Clone repository:

[https://github.com/RolandGuijt/
react-fundamentals](https://github.com/RolandGuijt/react-fundamentals)

`npm install`

`npm run dev`



ESLint

Detect problems

Code styling

Commonly used

Vite: Built-in React ruleset



Next up:
Server-side React
and Next.js

